30

・定積分を最小にする定数の決定

(例) 定積分 $\int_{-\pi}^{\pi}\{x-(a\sin x+b\sin 2x)\}^2dx$ を最小にする実数 a、b の値を求めよ。

x, $\sin x$, $\sin 2x$ は ⊖

× =

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$$(与式)=\int_{-\pi}^{\pi}(x^2+a^2\sin^2 x+b^2\sin^2 2x-2ax\sin x-2bx\sin 2x+2ab\sin 2x\sin x)dx$$
$$=2\int_0^{\pi}(x^2+a^2\sin^2 x+b^2\sin^2 2x-2ax\sin x-2bx\sin 2x+2ab\sin 2x\sin x)dx$$

ここで

$$\int_0^{\pi}x^2dx=\left[\frac{1}{3}x^3\right]_0^{\pi}=\frac{1}{3}\pi^3$$

$$\int_0^{\pi}\sin^2 x\,dx=\int_0^{\pi}\frac{1-\cos 2x}{2}dx=\frac{1}{2}\left[x-\frac{1}{2}\sin 2x\right]_0^{\pi}=\frac{1}{2}\pi$$

$$\int_0^{\pi}\sin^2 2x\,dx=\int_0^{\pi}\frac{1-\cos 4x}{2}dx=\frac{1}{2}\left[x-\frac{1}{4}\sin 4x\right]_0^{\pi}=\frac{1}{2}\pi$$

$$\begin{aligned}\int_0^{\pi}x\sin x\,dx&=\int_0^{\pi}x(-\cos x)'dx\\&=\left[-x\cos x\right]_0^{\pi}+\int_0^{\pi}1\cdot\cos x\,dx\\&=\pi+\left[\sin x\right]_0^{\pi}\\&=\pi\end{aligned}$$

$$\begin{aligned}\int_0^{\pi}x\sin 2x\,dx&=\int_0^{\pi}x\left(-\frac{1}{2}\cos 2x\right)'dx\\&=\left[-\frac{1}{2}x\cos 2x\right]_0^{\pi}+\frac{1}{2}\int_0^{\pi}1\cdot\cos 2x\,dx\\&=-\frac{1}{2}\pi+\frac{1}{2}\left[\frac{1}{2}\sin 2x\right]_0^{\pi}\\&=-\frac{1}{2}\pi\end{aligned}$$

$$\begin{aligned}\int_0^{\pi}\sin 2x\sin x\,dx&=-\frac{1}{2}\int_0^{\pi}\{\cos(2x+x)-\cos(2x-x)\}dx\\&=-\frac{1}{2}\int_0^{\pi}(\cos 3x-\cos x)dx\\&=-\frac{1}{2}\left[\frac{1}{3}\sin 3x-\sin x\right]_0^{\pi}\\&=0\end{aligned}$$

であるから

$$\begin{aligned}(与式)&=2\left\{\frac{1}{3}\pi^3+a^2\cdot\frac{1}{2}\pi+b^2\cdot\frac{1}{2}\pi-2a\cdot\pi-2b\cdot\left(-\frac{1}{2}\pi\right)+2ab\cdot 0\right\}\\&=\pi a^2-4\pi a+\pi b^2+2\pi b+\frac{2}{3}\pi^3\\&=\pi(a-2)^2-4\pi+\pi(b+1)^2-\pi+\frac{2}{3}\pi^3\\&=\pi(a-2)^2+\pi(b+1)^2+\frac{2}{3}\pi^3-5\pi\end{aligned}$$

よって、

$a=2$、$b=-1$ のとき最小値 $\frac{2}{3}\pi^3-5\pi$